

Evaluation of Aloin and Aloe-Emodin as Anti-Inflammatory Agents in Aloe by Using Murine Macrophages

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The aloe ingredients responsible for physiological effects and the concentrations required to exert their biological activities are not fully understood. This study compares the anti-inflammatory effects of aloin and aloe-emodin with other polyphenols. Our results demonstrated that aloe-emodin dose-dependently inhibited inducible nitric oxide synthase (iNOS) mRNA expression and nitric oxide (NO) production at 5–40 μ M. In addition, the levels of cyclooxygenase-2 (COX-2) mRNA and prostaglandin E₂ (PGE₂) production were suppressed by 40 μ M aloe-emodin. Aloin also suppressed the production of NO at 5–40 μ M, although it did not suppress PGE₂ production. The present results indicate that aloin and aloe-emodin possibly suppress the inflammatory responses by blocking iNOS and COX-2 mRNA expression. The anti-inflammatory effect of aloe-emodin was comparable to that of kaempferol and quercetin, indicating aloe-emodin as a possible key constituent responsible for the anti-inflammatory activity of aloe.

Key words: aloin; aloe-emodin; polyphenol; anti-inflammatory effect

Aloe plants have been used in folk medicine as the remedy for a variety of conditions. Aloin, also called barbaloin, is a bitter-tasting yellow crystal, and is the C-glycoside derivative of an anthraquinone.¹⁾ Orally administered, aloin is known to be hydrolyzed by the esterases secreted by intestinal microflora.^{2,3)} Once the C-glycoside has been hydrolyzed, it forms the aloe-emodin, anthrone which is further auto-oxidized to the quinone, aloe-emodin. Aloin and aloe-emodin possess not only a laxative effect, but antibacterial, antiviral, hepatoprotective, and anticancer effects.^{4–7)} Aloin and aloe-emodin are the major anthraquinones in aloe plants, and it has been noted that the level of these anthraquinones ranges from 0.1 to 21.5% dry weight in the leaf exudate of 68 aloe species.^{8,9)} Since aloin and aloe-emodin contain a polyphenolic structure, these compounds may also be responsible for the reported anti-inflammatory effects of aloe.^{10,11)}

Chronic inflammation is involved in the pathogenesis of many chronic diseases, including inflammatory bowel disease, rheumatoid arthritis, sepsis, and cancer.^{12–15)} Therefore, the suppressing the production of pro-inflammatory molecules could be an important target for the prevention or treatment of various diseases. The

LPS-treated RAW 264.7 murine macrophage model is widely used to study inflammatory responses. Exposure of RAW 264.7 macrophages to external bacterial toxins such as lipopolysaccharides (LPS) stimulates the secretion of pro-inflammatory mediators including nitric oxide (NO) and prostaglandin E₂ (PGE₂) which are respectively produced by the inducible isoforms of nitric oxide synthase (iNOS) and cyclooxygenase-2 (COX-2).¹⁶⁾

In the present study, as a prelude to reveal the underlying mechanism for the anti-inflammatory effect of aloe, we evaluated the effects of aloin and aloe-emodin against LPS-stimulated NO and PGE₂ release in RAW 264.7 macrophages. Moreover, we compared their activity to other polyphenols which are known to exhibit anti-inflammatory effects.

Materials and Methods

Chemicals. Dulbecco's modified Eagle's medium (DMEM), fetal bovine serum (FBS), sodium pyruvate, L-glutamine, an anti-biotics-antimycotics solution, and trypsin-EDTA were purchased from GIBCO Co. (Grand Island, NY, USA). Lipopolysaccharide (LPS, *Escherichia coli* 0111: B4) and other chemicals otherwise indicated were from Sigma (St. Louis, MO, USA).

Cell culture. Mouse macrophage RAW 264.7 macrophages, obtained from the Korean Cell Line Bank (KCLB, Seoul, Korea), were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% heat-inactivated FBS, 100 units/ml of penicillin, and 100 μ g/ml of streptomycin. The cells were incubated at 37 °C and 5% CO₂ in a humidified cell incubator. For all assays, the RAW 264.7 macrophages were plated in 60-mm culture dishes (3×10^6 cells) for 12 h, before being treated with LPS (1 μ g/ml) and different concentrations (5, 10, and 40 μ M) of the test compounds, including anthraquinones (aloin and aloe-emodin), two isoflavones (genistein and daidzein), and two flavonols (kaempferol and quercetin), for a further 24 h.

Cell viability. After 24 h of incubation, the macrophages were washed in PBS, detached from the culture dishes by exposure to trypsin-EDTA, and resuspended in the culture medium. The resulting macrophages were then stained with trypan blue at a final concentration of 0.2%, after which the trypan blue-excluded viable cells were counted with a hemacytometer under a microscope.

Determination of NO and PGE₂. After the treatment, the medium was collected and centrifuged at 2000 $\times g$ for 10 min to obtain a cell-free fraction. The supernatant was decanted into a new microcentrifuge tube, and the amount of NO and PGE₂ determined. The amount of NO production was measured in the culture medium by using a commercially available NO detection kit (iNtRON, Seoul, Korea) according to the manufacturer's protocol. This assay determines the

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nitric oxide (NO) concentration based on the enzymatic conversion of nitrate to nitrite by nitrate reductase. The reaction was followed by the colorimetric detection of nitrite as an azo dye product of the Griess reaction. Briefly, 100 μ l of the medium was mixed with 50 μ l of Griess reagent I and 50 μ l of Griess reagent II in a 96-well plate. After 10 min, the optical density was read in a microplate reader at 540 nm. The PGE₂ concentration in the supernatant of the culture medium was determined by using a commercially available PGE₂ competitive enzyme immunoassay kit (Amersham Pharmacia Biotech., Piscataway, NJ, USA) according to the manufacturer's protocol. This assay is based on the competitive binding technique in which PGE₂ present in a sample competes with a fixed amount of horseradish peroxidase-labeled PGE₂ for sites on a mouse monoclonal antibody. During the incubation, the mouse monoclonal antibody becomes bound to the goat anti-mouse antibody coated on the microplate. Briefly, 100 μ l of the medium was mixed with 50 μ l of a mouse monoclonal antibody solution and 50 μ l of a PGE₂ conjugate (horseradish peroxidase) in a 96-well plate. After 2 h, the optical density was read in a microplate reader at 450 nm.

RNA isolation and reverse transcriptase-polymerase chain reaction (RT-PCR). To determine the time-course characteristics of iNOS and COX-2 mRNA expression in RAW 264.7 macrophages exposed to aloin or aloe-emodin, macrophages were treated with aloin (40 μ M) or aloe-emodin (40 μ M) for 6, 12 or 24 h. To investigate the changes of iNOS and COX-2 mRNA expression induced by polyphenols, RT-PCR was performed. RAW 264.7 macrophages were treated with each polyphenol (5, 10, and 40 μ M) for 24 h and then washed with ice-cold PBS. Total RNA was isolated by using TRIZOL (Gibco BRL Life Technologies., Grand Island, NY, USA), and the total RNA concentration was quantified by spectrophotometry at 260 nm. Total RNA (2 μ g) was converted to cDNA with an oligo d(T) primer. PCR was performed on the cDNA by using the following sense and antisense primers. The PCR primers used in this study are listed next and were purchased from Bioneer (Daejeon, Korea). The sense primer for iNOS was 5'-AATGGCAACATCAGGTCGGCCATCACT-3', and the antisense primer was 5'-GCTGTGTGTACAGAAAGTCTCGAACTC-3'. The sense primer for COX-2 was 5'-GGAGAGACTATCAAGATAGT-3', and the antisense primer was 5'-ATGGTCAGTAGACTTTTACA-3'.¹⁷⁾ The sense primer for GAPDH was 5'-CCATCAATGACCCCTTCATTGACC-3', and the antisense primer was 5'-GAAGGCCATGCCAGTGAGCTTCC-3'.¹⁸⁾ PCR of the cDNA was performed in a final volume of 20 μ l containing a PCR primer, oligo d(T), total RNA, and DEPC-H₂O. Each PCR step was performed as follows: initial denaturation at 94 °C for 5 min; 25 cycles of amplification consisting of denaturation at 94 °C for 30 s, annealing at 60 °C for 30 s, elongation at 72 °C for 1 min, and extension at 72 °C for 5 min. The PCR products were run on a 2% agarose gel and visualized by ethidium bromide staining.

Statistical analysis. All the data from the experiment are expressed as the mean $\pm$ S.D. Data were analyzed by a one-way analysis of variance (ANOVA) and followed by Duncan's multiple-range test. Differences are considered statistically significant at $p < 0.05$.

Results

Cell viability

The trypan blue dye exclusion assay showed that RAW 264.7 macrophages treated with a test compound for 24 h maintained >90% of viability. The viability of the aloin (5, 10, and 40 μ M)-treated cells was 94.3%, 96.1%, and 91.2%, respectively. The aloe-emodin treatment (5, 10, and 40 μ M) showed 95.1%, 93.0%, and 90.9% maintenance, respectively.

Effects of anthraquinones, isoflavones, and flavonols on the NO production

The chemical structures of the two anthraquinones (aloin and aloe-emodin), two isoflavones (genistein and

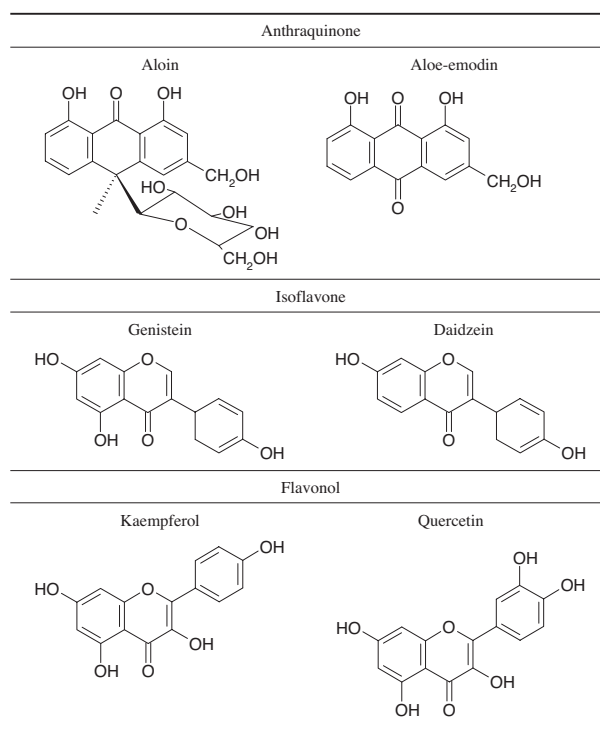


Fig. 1. Chemical Structures of the Test Compounds.

Table 1. Nitric Oxide (NO) Production in RAW 264.7 Macrophages Stimulated with LPS (1 μ g/ml) in the Presence of Test Compounds (5–40 μ M)

Agent	Conc. (μ mole/l)		
	5 μ M	10 μ M	40 μ M
LPS (–)		0.51 $\pm$ 0.41 ^f	
LPS (+)		9.94 $\pm$ 0.56 ^a	
LPS + Aloin	8.45 $\pm$ 0.48 ^{bc}	8.78 $\pm$ 0.20 ^b	8.19 $\pm$ 0.62 ^b
LPS + Aloe-emodin	5.95 $\pm$ 0.84 ^{cd}	5.49 $\pm$ 0.27 ^d	3.25 $\pm$ 0.29 ^d
LPS + Genistein	8.75 $\pm$ 0.14 ^b	7.02 $\pm$ 1.18 ^c	4.28 $\pm$ 0.89 ^{cd}
LPS + Daidzein	7.68 $\pm$ 1.24 ^{bc}	6.40 $\pm$ 0.43 ^{cd}	4.39 $\pm$ 1.43 ^c
LPS + Kaempferol	6.39 $\pm$ 0.82 ^c	4.66 $\pm$ 0.18 ^c	2.86 $\pm$ 0.25 ^e
LPS + Quercetin	4.66 $\pm$ 0.69 ^d	4.43 $\pm$ 0.22 ^e	2.82 $\pm$ 0.56 ^e

RAW 264.7 macrophages were stimulated with 1 μ g/ml of LPS in the absence or presence of polyphenols (5, 10, 40 μ M). After 24 h of incubation, the medium was collected and the NO production measured. The LPS (–) and LPS (+) values were obtained in the absence of a polyphenol (0 μ M). Each value is the mean $\pm$ SD, $n = 4$. Means in a row with different superscripts are significantly different ($p < 0.05$).

daidzein), and two flavonols (kaempferol and quercetin) used in the present study are shown in Fig. 1. The LPS-stimulated NO production in RAW 264.7 macrophages was investigated. All the test compounds showed dose-dependent suppression of NO release, except for aloin. The potency of inhibition was in the order of quercetin $\geq$ aloe-emodin $\geq$ kaempferol $\geq$ daidzein $\geq$ genistein $\geq$ aloin at the lowest concentration (5 μ M) (Table 1).

Effects of anthraquinones, isoflavones, and flavonols on the iNOS mRNA expression

First, we determined the time-course characteristics of iNOS and COX-2 mRNA expression in RAW 264.7 macrophages exposed to aloin (40 μ M) or aloe-emodin (40 μ M). LPS (1 μ g/ml)-treated macrophages showed time-dependent increases in iNOS and COX-2 mRNA

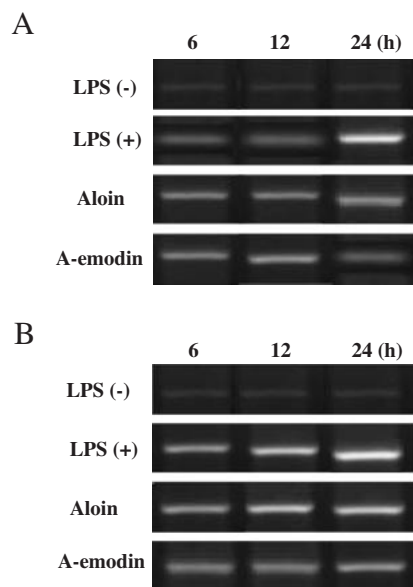


Fig. 2. Time Course Characteristics of iNOS (A) and COX-2 (B) mRNA Expression in RAW 264.7 Macrophages Exposed to Aloin or Aloe-Emodin.

RAW 264.7 macrophages were stimulated with 1 µg/ml of LPS in the absence or presence of aloe polyphenols (40 µM). The cells were incubated with aloin or aloe-emodin for 6, 12, and 24 h.

Table 2. Prostaglandin E₂ (PGE₂) Production in RAW 264.7 Macrophages Stimulated with LPS (1 µg/ml) in the Presence of Test Compounds (5–40 µM)

Agent	Conc. (µmole/l)		
	5 µM	10 µM	40 µM
LPS (–)			
LPS (+)		147.5 ± 68.8 ^d	
LPS + Aloin	1116.6 ± 30.6 ^{ab}	1068.3 ± 22.1 ^a	966.8 ± 71.2 ^{ab}
LPS + Aloe-emodin	1090.9 ± 45.2 ^{ab}	1031.5 ± 60.7 ^{ab}	460.7 ± 42.8 ^c
LPS + Genistein	1103.2 ± 40.3 ^{ab}	1065.7 ± 33.7 ^{ab}	1109.7 ± 19.8 ^{ab}
LPS + Daidzein	1012.5 ± 40.1 ^b	1032.1 ± 30.9 ^{ab}	975.7 ± 50.2 ^b
LPS + Kaempferol	946.1 ± 24.9 ^c	925.9 ± 75.4 ^b	419.8 ± 73.7 ^c
LPS + Quercetin	963.3 ± 103.6 ^{bc}	811.0 ± 31.7 ^c	445.3 ± 8.3 ^c

RAW 264.7 macrophages were stimulated with 1 µg/ml of LPS in the absence or presence of a polyphenol (5, 10, 40 µM). After 24 h of incubation, the medium was collected and the PGE₂ production measured. The LPS (–) and LPS (+) values were obtained in the absence of a polyphenol (0 µM). Each value is the mean ± SD, n = 4. Means in a row with different superscripts are significantly different (p < 0.05).

expression during the 24 h incubation period, and the test compounds suppressed the expression at each incubation time point (Fig. 2). We next examined the expression of iNOS mRNA in macrophages treated with different concentrations of each compound for 24 h. At a high concentration (40 µM), all the compounds inhibited the LPS-induced up-regulation of iNOS (Fig. 3). However, only aloe-emodin, quercetin, and kaempferol were effective at the lower concentrations.

Effects of anthraquinones, isoflavones, and flavonols on the PGE₂ production

The PGE₂ level was markedly increased (784.20 ± 44.2%) after 24 h of LPS (1 µg/ml) treatment. Aloe-emodin effectively reduced the level of PGE₂ at a 40 µM

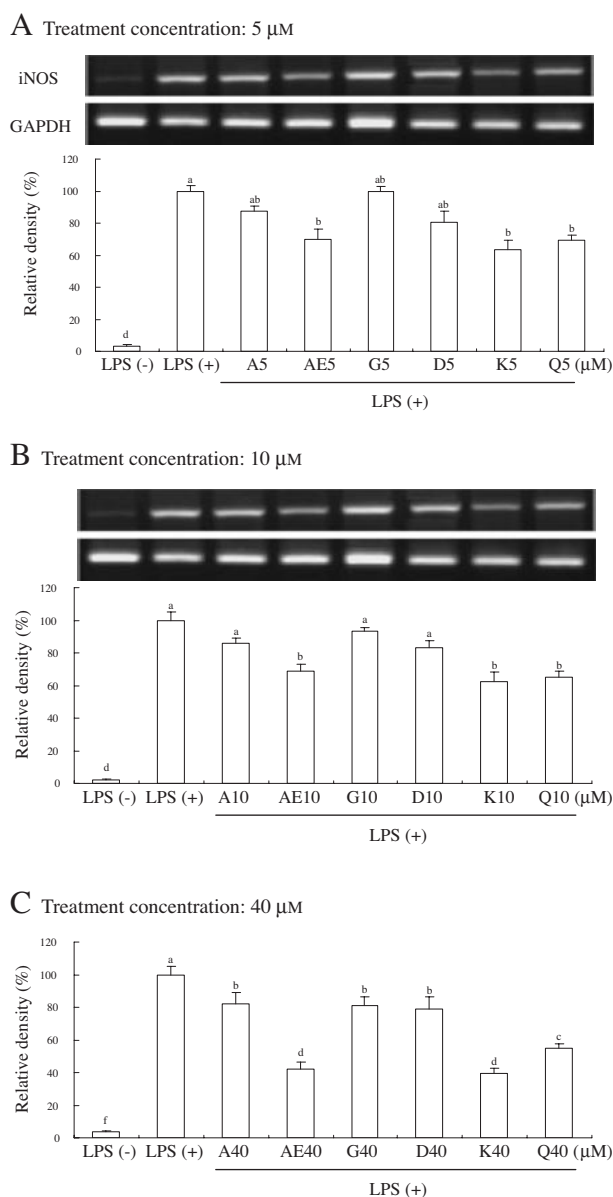


Fig. 3. Expression of Inducible Nitric Oxide Synthase (iNOS) mRNA in RAW 264.7 Macrophages Stimulated with LPS (1 µg/ml) in the Presence of Test Compounds (5–40 µM).

RAW 264.7 macrophages were stimulated with 1 µg/ml of LPS in the absence or presence of a polyphenol. A, aloin; AE, aloe-emodin; G, genistein; D, daidzein; K, kaempferol; Q, quercetin. A, The cells were treated with 1 µg/ml of LPS and 5 µM of a test compound for 24 h (n = 4). B, The cells were treated with 1 µg/ml of LPS and 10 µM of a test compound for 24 h (n = 4). C, The cells were treated with 1 µg/ml of LPS and 40 µM of a test compound for 24 h (n = 4). Each bar represents the mean ± S.D. The bars with different letters are significantly different (p < 0.05) from each other.

concentration. However, aloin did not show a significant effect at the concentrations used in this study. Kaempferol and quercetin showed significantly suppressed PGE₂ production at all concentrations used in a dose-dependent manner. Daidzein and genistein did not reduce the level of PGE₂.

Effects of anthraquinones, isoflavones, and flavonols on the COX-2 mRNA expression

To investigate whether the inhibition of PGE₂ production was due to modulation of the COX-2 gene level, we performed an RT-PCR analysis of COX-2

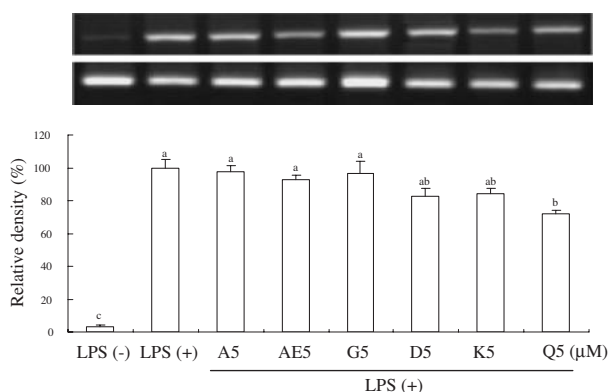
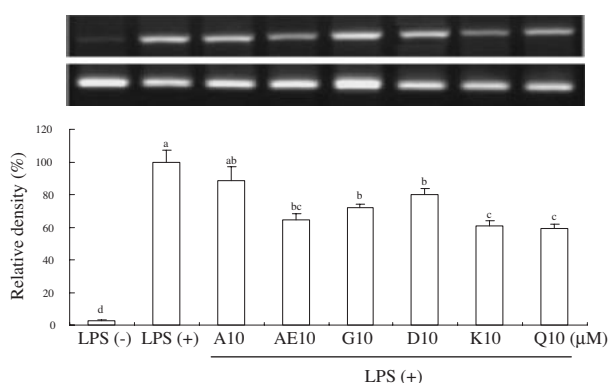
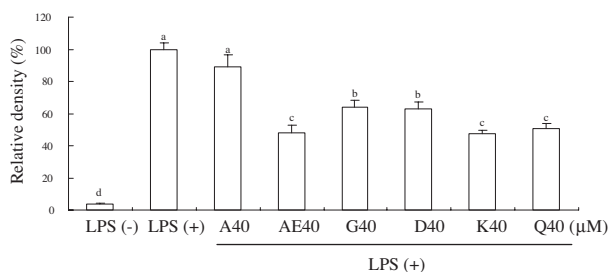
A Treatment concentration: 5 μ MB Treatment concentration: 10 μ MC Treatment concentration: 40 μ M

Fig. 4. Expression of Cyclooxygenase-2 (COX-2) mRNA in RAW 264.7 Macrophages Stimulated with LPS (1 μ g/ml) in the Presence of Test Compounds (5–40 μ M).

RAW 264.7 macrophages were stimulated with 1 μ g/ml of LPS in the absence or presence of a polyphenol. A, aloin; AE, aloe-emodin; G, genistein; D, daidzein; K, kaempferol; Q, quercetin. A, The cells were treated with 1 μ g/ml of LPS and 5 μ M of a test compound for 24 h ($n = 4$). B, The cells were treated with 1 μ g/ml of LPS and 10 μ M of a test compound for 24 h ($n = 4$). C, The cells were treated with 1 μ g/ml of LPS and 40 μ M of a test compound for 24 h ($n = 4$). Each bar represents the mean $\pm$ S.D. The bars with different letters are significantly different ($p < 0.05$) from each other.

mRNA expression. The LPS-induced COX-2 mRNA expression was inhibited by aloe-emodin in a dose-dependent manner (Fig. 4). This inhibition of COX-2 mRNA was correlated to the inhibition of PGE₂ level by aloe-emodin. However, aloin did not affect the LPS-induced COX-2 mRNA expression as expected. Both genistein and daidzein suppressed the COX-2 mRNA expression at 10 and 40 μ M concentrations. The COX-2 mRNA expression was significantly inhibited by kaempferol and quercetin, which coincides with the PGE₂ results.

Discussion

Aloe contains several active compounds, including aloin, a C-glycoside that can be hydrolyzed in the gut to form aloe-emodin anthrone which, in turn, is auto-oxidized to the quinone aloe-emodin. Aloin and aloe-emodin have been pointed out as possible active compounds to relieve a variety of ailments related to chronic inflammation. However, few studies have investigated the relative biological activity of aloin and aloe-emodin.

The results of the present study indicate that the aloe anthraquinones inhibited the LPS-induced inflammatory response in RAW 264.7 macrophages. The activity of aloe-emodin was comparable to that of kaempferol and to quercetin which are known potent inhibitors of inflammation.

We used macrophages to evaluate the anti-inflammatory activity, since it is well known that macrophages play a crucial role in both non-specific and acquired immune responses. The activation of macrophages by LPS leads to a functionally diverse series of responses including the activation of phospholipase A₂, which produces lipid metabolites of arachidonic acid such as prostaglandins, and the production of NO, and to the production of pro-inflammatory cytokines (IL-1, TNF- α and IL-6).¹⁹⁾

The enzyme, cyclooxygenase-2 (COX-2), which catalyzes prostaglandin (PG) biosynthesis, has become an important target for the discovery and development of new anti-inflammatory agents. Flavonoids with a free hydroxyl group at the 3- or 7-position are known to act as antioxidants or to possess anti-inflammatory properties.^{20,21)} This is consistent with our previous observation that quercetin possessed a strong inhibitory effect on COX-2 activity.²²⁾ A structure-activity study has indicated that more than 2 hydroxyl groups on the B ring and the presence of the oxo group at the 4-position of the C ring were important for the suppression of COX-2 transcription activity.²³⁾ Aloe-emodin is not a flavonoid, while possessing two hydroxyl groups, and its anti-inflammatory activity was similar to that of kaempferol and quercetin which were strong inhibitors in the present study. A previous report has suggested that the position of the hydroxyl groups on one or both sides of the anthraquinone structure significantly affected the radical scavenging activity of hydroxyanthraquinones. The hydroxy groups of aloe-emodin molecules are on both rings of the anthraquinone structure which is not an *ortho*-dihydroxy structure. This type of hydroxyanthraquinone structure has shown poor antioxidative activity in an ABT^{•+} assay²⁴⁾ and also in a DPPH assay.²⁵⁾ Therefore, the anti-inflammatory activity of aloe-emodin may not be caused by its antioxidative activity. Other investigators have pointed out that the exposure of cancer cells to aloe-emodin could lead to DNA damage^{26,27)} and induce apoptosis which is mediated through the activation of p53 and caspase-3.²⁸⁾ The apoptotic process induced by aloe-emodin is further amplified by caspase itself to down-regulate NF- κ B.²⁹⁾ Therefore, it is possible that the susceptibility to COX-2 inhibitory effects of aloe-emodin may be correlated with modulation of the NF- κ B signaling pathway.

The data provided in the present report indicate that both aloin and aloe-emodin effectively reduced NO

release, although only aloe-emodin significantly reduced iNOS expression. There are several transcription factor binding sites located on the promoter of the murine iNOS gene such as IFN- γ response elements (IRF), the γ -activated site (GAS), Oct-1, and NF- κ B.^{30–32} The exact mechanism underlying the inhibitory action of aloe-emodin toward iNOS expression in macrophages is not known. However, emodin has been reported to inhibit the NF- κ B gene expression in macrophages.^{33,34} Aloe-emodin has also been shown to block the activation of IRF-1, but not NF- κ B in cancer cells.³⁵

We have shown that aloin, a glycoside, did not affect COX-2 or iNOS mRNA expression nor PGE₂ production. The glycosylation of flavonoids has reduced their activity when compared to the corresponding aglycones.³⁶ In our study, glycosylation of the hydroxyanthraquinone, aloin, diminished its anti-inflammatory activity due to the stereo-shading effect of the bulky structure.

Isoflavones (genistein and daidzein) were weak anti-inflammatory agents compared to flavonols (kaempferol and quercetin). This finding is consistent with an other report that kaempferol and quercetin were more effective in the suppression of NO production than genistein and genistin³⁷ and that flavonols were stronger collagenase inhibitors than the flavones of skin inflammation.³⁸ The authors suggested that C-3 hydroxyl substitution was important to the anti-inflammatory responses.

The results of the present study indicate that aloin and aloe-emodin possessed anti-inflammatory activity and did not affect murine macrophage viability. In particular, aloe-emodin was an effective inhibitor of inflammatory responses, and its activity was similar to that of the potent flavonols. The *in vivo* anti-inflammatory activity of aloe-emodin is currently under investigation.

Acknowledgments

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References

- 1) Saccù D, Bogoni P, and Procida G, *J. Agric. Food Chem.*, **49**, 4526–4530 (2001).
- 2) Hattori M, Kanda T, Shu YZ, Akao T, Kobashi K, and Namba T, *Chem. Pharm. Bull. (Tokyo)*, **36**, 4462–4466 (1988).
- 3) Che QM, Akao T, Hattori M, Kobashi K, and Namba T, *Planta Med.*, **57**, 15–19 (1991).
- 4) Hatano T, Kusuda M, Inada K, Ogawa TO, Shiota S, Tsuchiya T, and Yoshida T, *Phytochemistry*, **66**, 2047–2055 (2005).
- 5) Andersen DO, Weber ND, Wood SG, Hughes BG, Murray BK, and North JA, *Antiviral Res.*, **16**, 185–196 (1991).
- 6) Arosio B, Gagliano N, Fusaro LM, Parmeggiani L, Tagliabue J, Galetti P, De Castri D, Moscheni C, and Annoni G, *Pharmacol. Toxicol.*, **87**, 229–233 (2000).
- 7) Esmat AY, Tomasetto C, and Rio MC, *Cancer Biol. Ther.*, **5**, 97–103 (2006).
- 8) Groom QJ and Reynolds T, *Planta Med.*, **53**, 345–348 (1987).
- 9) van Wyk BE, van Rheede, van Oudtshoorn MC, and Smith GF, *Planta Med.*, **61**, 250–253 (1995).
- 10) Somboonwong J, Thanamitramanee S, Jariyapongskul A, and Patumraj S, *J. Med. Assoc. Thai.*, **83**, 417–425 (2000).
- 11) Korkina L, Suprun M, Petrova A, Mikhal'chik E, Luci A, and DeLuca C, *Biofactors*, **18**, 255–264 (2003).
- 12) Ljung T, Lundberg S, Varsanyi M, Johansson C, Schmidt PT, Herulf M, Lundberg JO, and Hellstrom PM, *World J. Gastroenterol.*, **7**, 3386–3392 (2006).
- 13) Sakaguchi Y, Shirahase H, Ichikawa A, Kanda M, Nozaki Y, and Uehara Y, *Life Sci.*, **24**, 2257–2267 (2004).
- 14) Huynh T, Lemasters JJ, Bracey LW, and Baker CC, *Shock*, **14**, 555–560 (2000).
- 15) Watanabe K, Kawamori T, Nakatsugi S, and Wakabayashi K, *Biofactors*, **12**, 129–133 (2000).
- 16) Lee SH, Soyoola E, Chanmugam P, Hart S, Sun W, Zhong H, Liou S, Simmons D, and Hwang D, *J. Biol. Chem.*, **25**, 25934–25938 (1992).
- 17) Kim IT, Park YM, Won JH, Park HJ, Choi JW, and Lee KT, *Biol. Pharm. Bull.*, **28**, 94–100 (2005).
- 18) Chiang YM, Lo CP, Chen YP, Wang SY, Yang NS, Kuo YH, and Shyur LF, *Br. J. Pharmacol.*, **146**, 352–363 (2005).
- 19) Heiss E, Herhaus C, Klimo K, Bartsch H, and Gerhauser C, *J. Biol. Chem.*, **276**, 32008–32015 (2001).
- 20) Sartor L, Pezzato E, Dell'Aica I, Caniato R, Biggin S, and Garbisa S, *Biochem. Pharmacol.*, **64**, 229–237 (2002).
- 21) Gil B, Sanz MJ, Terencio MC, Ferrandiz ML, Bustos G, Paya M, Gunasegaran R, and Alcaraz MJ, *Life Sci.*, **54**, PL333–PL338 (1994).
- 22) Jung WJ and Sung MK, *Biofactors*, **21**, 113–117 (2004).
- 23) Mutoh M, Takahashi M, Fukuda K, Matsushima-Hibiya Y, Mutoh H, Sugimura T, and Wakabayashi K, *Carcinogenesis*, **21**, 959–963 (2000).
- 24) Cai Y, Sun M, Xing J, and Corke H, *J. Agric. Food Chem.*, **29**, 7884–7890 (2004).
- 25) Matsuda H, Morikawa T, Toguchida I, Park JY, Harima S, and Yoshikawa M, *Bioorg. Med. Chem.*, **9**, 41–50 (2001).
- 26) Lee HZ, Hsu SL, Liu MC, and Wu CH, *Eur. J. Pharmacol.*, **23**, 287–295 (2001).
- 27) Kuo PL, Lin TC, and Lin CC, *Life Sci.*, **6**, 1879–1892 (2002).
- 28) Lin JG, Chen GW, Li TM, Chouh ST, Tan TW, and Chung JG, *J. Urol.*, **175**, 343–347 (2006).
- 29) Chen HC, Hsieh WT, Chang WC, and Chung JG, *Food Chem. Toxicol.*, **42**, 1251–1257 (2004).
- 30) Kamijo R, Harada H, Matsuyama T, Bosland M, Gerecitano J, Shapiro D, Le J, Koh SI, Kimura T, and Green SJ, *Science*, **263**, 1612–1615 (1994).
- 31) Vila-del Sol V, Díaz-Muñoz MD, and Fresno M, *J. Leukoc. Biol.*, **81**, 272–283 (2007).
- 32) Kim YM, Ko CB, Park YP, Kim YJ, and Paik SG, *Mol. Cells*, **9**, 99–109 (1999).
- 33) Li HL, Chen HL, Li H, Zhang KL, Chen XY, Wang XW, Kong QY, and Liu J, *Int. J. Mol. Med.*, **16**, 41–47 (2005).
- 34) Chen Y, Yang L, and Lee TJ, *Biochem. Pharmacol.*, **59**, 1445–1457 (2000).
- 35) Mijatovic S, Maksimovic-Ivanic D, Radovic J, Popadic D, Momcilovic M, Harhaji L, Miljkovic D, and Trajkovic V, *Cell. Mol. Life Sci.*, **61**, 1805–1815 (2004).
- 36) Rice-Evans CA, Miller NJ, and Paganga G, *Free Radic. Biol. Med.*, **20**, 933–956 (1996).
- 37) Wang J and Mazza G, *J. Agric. Food Chem.*, **50**, 850–857 (2002).
- 38) Sin BY and Kim HP, *Arch. Pharm. Res.*, **28**, 1152–1155 (2005).